

EvansFaceId on Raspberry Pi 3

Overview

This project runs a **face recognition and logging system** on a Raspberry Pi 3 using a camera module. The system detects faces, compares them against known encodings (`.npy` files) for multiple people, and logs detections to a file with a time buffer to prevent repeated entries.

This version uses a **minimum-distance comparison** for unbiased recognition, scales camera frames for performance, and is optimized for the Raspberry Pi 3.

Requirements

- Raspberry Pi 3 (1 GB RAM)
- Raspberry Pi Camera Module (v1 or v2)
- microSD card with **Raspberry Pi OS 32-bit**
- Python 3.7–3.9 recommended
- `pip` and `venv` for virtual environments

Optional: Monitor or SSH for headless operation.

Hardware Setup

1. Connect your Raspberry Pi Camera Module to the camera port (CSI) on the Pi.
2. Enable the camera in Raspberry Pi settings:

```
sudo raspi-config
```

3. Go to **Interface Options → Camera → Enable**
4. Reboot when prompted.
5. Test the camera:

```
libcamera-hello
```

6. You should see a live preview. Press Ctrl+C to exit.
7. Optional: Capture a test image:

```
libcamera-jpeg -o test.jpg
```

Software Setup

Step 1: Update and upgrade the Pi

```
sudo apt update && sudo apt full-upgrade -y  
sudo reboot
```

Step 2: Install Python and dependencies for building packages

```
sudo apt install -y python3 python3-venv python3-pip build-essential cmake  
libatlas-base-dev libopenblas-dev libjpeg-dev libboost-all-dev
```

Step 3: Create a Python virtual environment

```
python3 -m venv ~/faceid-env  
source ~/faceid-env/bin/activate
```

- The `source` command activates the environment.
- All subsequent installations will be **isolated** to this environment.

Step 4: Upgrade pip inside the virtual environment

```
pip install --upgrade pip
```

Step 5: Install required Python libraries

```
pip install numpy opencv-python face-recognition
```

Note: On Raspberry Pi 3, building `dlib` (a dependency of `face-recognition`) may take a long time (30–60 min).

Step 6: Transfer the project files

Copy the following files to your Pi (e.g., via `scp` or USB):

- `EvansFaceId.py` (the updated script)

- `known_faces/` folder containing `.npy` encodings for each person

Example:

```
scp -r EvansFaceId.py known_faces pi@<pi_ip>:/home/pi/
```

Running the Script

1. Activate the virtual environment:

```
source ~/faceid-env/bin/activate
```

2. Navigate to the project folder:

```
cd /home/pi/
```

3. Run the script:

```
python EvansFaceId.py
```

4. Press `q` to quit.
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Performance Tips for Raspberry Pi 3

- Reduce frame size for faster processing:

```
small_frame = cv2.resize(frame, (0, 0), fx=0.2, fy=0.2)
```

- Increase `compare_interval` to 1-2 seconds for stability:

```
face_id.compare_interval = 1.0
```

- Raise `FaceTolerance` to ~0.5-0.6 if recognition is inconsistent:

```
face_id.FaceTolerance = 0.5
```

- For headless operation, comment out `cv2.imshow()` calls.
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Sending Logs to a Computer Over the Internet

There are multiple options to transfer the log file remotely:

Option 1: SCP/SFTP

- From your remote computer:

```
scp pi@<pi_ip>:/home/pi/face_log.txt ./face_log.txt
```

- You can set up a cron job on the Pi to automatically copy the log every hour.

Option 2: Upload via FTP/SFTP

- Install an FTP server on the Pi (`vsftpd`) and configure remote access.

Option 3: Cloud Storage

- Use Python to upload the log file to Google Drive, Dropbox, or AWS S3. Example with Dropbox:

```
import dropbox
dbx = dropbox.Dropbox('<ACCESS_TOKEN>')
with open('/home/pi/face_log.txt', 'rb') as f:
    dbx.files_upload(f.read(), '/face_log.txt',
mode=dropbox.files.WriteMode.overwrite)
```

- Schedule a cron job to upload periodically.

Option 4: Email

- Use `smtplib` in Python to email the log file periodically.
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Optional: Auto-start on boot (systemd service)

1. Create a service file:

```
sudo nano /etc/systemd/system/faceid.service
```

Paste:

```
[Unit]
Description=Evans Face ID
After=network.target

[Service]
ExecStart=/home/pi/faceid-env/bin/python /home/pi/EvansFaceId.py
WorkingDirectory=/home/pi
StandardOutput=append:/home/pi/faceid.log
StandardError=append:/home/pi/faceid.log
Restart=always

[Install]
WantedBy=multi-user.target
```

1. Enable and start the service:

```
sudo systemctl enable faceid
sudo systemctl start faceid
```

2. Logs will be written to `/home/pi/faceid.log`.
3. The script will automatically restart if the Pi reboots.

Notes

- Store all known face encodings in `known_faces/<person_name>/` as `.npy` files.
- The script logs detected faces every few seconds (controlled by `log_buffer_time`).
- Adjust parameters in the script to balance **performance and recognition accuracy** for Raspberry Pi 3.

References

- [face_recognition GitHub](#)
- [OpenCV Python Documentation](#)
- [Raspberry Pi Camera Setup](#)